

Phase 1 — Linux, Networking & DevOps Fundamental

1. Navigation & File System Basics

pwd

Purpose:

Displays the current working directory.

```
pwd
```

ls

Purpose:

Lists files and directories in the current directory.

```
ls ls
-l ls
-a ls
-la ls
-lh
```

cd

Purpose:

Changes the current working directory.

```
cd /home
cd .. cd
~ cd -
```

mkdir

Purpose:

Creates a new directory.

```
mkdir project mkdir -p
project/src/config
```

touch

Purpose:

Creates an empty file or updates the timestamp of an existing file.

```
touch file.txt
```

cp

Purpose:

Copies files or directories.

```
cp file.txt backup.txt cp -r
project backup_project
```

mv

Purpose:

Moves or renames files and directories.

```
mv old.txt new.txt mv  
file.txt /home/user/
```

rm

Purpose:

Removes files or directories.

```
rm file.txt rm  
-r directory
```

rmdir

Purpose:

Removes an empty directory.

```
rmdir directory_name
```

file

Purpose:

Displays information about the type of a file.

```
file example.txt
```

stat

Purpose:

Displays detailed information about a file or directory, including permissions, ownership, size, and timestamps.

```
stat file.txt
```

tree

Purpose:

Displays files and directories in a tree-like structure.

```
tree
```

Note: The command may need to be installed separately.

basename

Purpose:

Extracts the filename from a path.

```
basename /home/user/file.txt
```

dirname

Purpose:

Extracts the directory portion from a path.

```
dirname /home/user/file.txt
```

ln

Purpose:

Creates hard links or symbolic links.

```
ln -s /path/to/file linkname
```

readlink

Purpose:

Displays the target of a symbolic link.

```
readlink linkname
```

2. Viewing & Processing File Content

cat

Purpose:

Displays the contents of a file.

```
cat file.txt
```

less

Purpose:

Views the contents of a file one screen at a time.

```
less file.txt
```

Press Q to exit.

head

Purpose:

Displays the beginning of a file.

```
head file.txt head  
-n 20 file.txt
```

tail

Purpose:

Displays the end of a file.

```
tail file.txt tail -f  
/var/log/syslog
```

grep

Purpose:

Searches for a specific text pattern inside files or command output.

```
grep "error" logfile.txt  
grep -i "error" logfile.txt  
grep -r "error" /var/log/
```

find

Purpose:

Searches for files and directories.

```
find /home -name "*.txt"  
find /home -type f find  
/home -type d
```

wc

Purpose:

Counts lines, words, and characters in a file.

```
wc file.txt wc  
-l file.txt
```

sort

Purpose:

Sorts lines of text.

```
sort names.txt
```

uniq

Purpose:

Removes or identifies repeated adjacent lines.

```
sort names.txt | uniq
```

cut

Purpose:

Extracts specific sections or columns from each line of text.

```
cut -d ":" -f 1 /etc/passwd
```

awk

Purpose:

Processes and extracts structured text and columns.

```
awk '{print $1}' file.txt
```

sed

Purpose:

Searches, replaces, and modifies text.

```
sed 's/old/new/g' file.txt
```

xargs

Purpose:

Builds and executes commands using input received from another command.

```
find . -name "*.log" | xargs grep "error"
```

tee

Purpose:

Displays command output and simultaneously writes it to a file.

```
command | tee output.txt  
command | tee -a output.txt
```

3. User & Group Management

whoami

Purpose:

Displays the username of the currently logged-in user.

```
whoami
```

id

Purpose:

Displays the user's UID, GID, and group memberships.

```
id id  
username
```

who

Purpose:

Shows users currently logged into the system.

```
who
```

w

Purpose:

Shows logged-in users and their current activities.

```
w
```

useradd

Purpose:

Creates a new user account.

```
sudo useradd username  
sudo useradd -m username
```

adduser

Purpose:

Creates a user account interactively.

```
sudo adduser username
```

passwd

Purpose:

Sets or changes a user's password.

```
sudo passwd username  
passwd
```

usermod

Purpose:

Modifies an existing user account.

```
sudo usermod -aG groupname username
```

userdel

Purpose:

Deletes a user account.

```
sudo userdel username  
sudo userdel -r username
```

groupadd

Purpose:

Creates a new group.

```
sudo groupadd groupname
```

groupdel

Purpose:

Deletes a group.

```
sudo groupdel groupname
```

groups

Purpose:

Displays the groups to which a user belongs.

```
groups groups  
username
```

su

Purpose:

Switches to another user account.

```
su username
```

sudo

Purpose:

Executes a command with administrative privileges according to the system's sudo policy.

```
sudo systemctl restart nginx
```

sudo -l

Purpose:

Displays the commands the current user is allowed to execute with sudo privileges.

```
sudo -l
```

passwd -S

Purpose:

Displays password status information for a user on systems that support this option.

```
sudo passwd -S username
```

last

Purpose:

Displays information about recent user login sessions.

```
last
```

lastlog

Purpose:

Displays the most recent login information for users.

```
lastlog
```

getent passwd

Purpose:

Displays user account information from the system's configured user database.

```
getent passwd
```

getent group

Purpose:

Displays group information from the system's configured group database.

```
getent group
```

4. Permissions & Ownership

chmod

Purpose:

Changes file or directory permissions.

```
chmod 644 file.txt  
chmod 755 script.sh  
chmod u+x script.sh  
chmod 640 sensitive.txt
```

chown

Purpose:

Changes the owner (and optionally group) of a file or directory.

```
sudo chown username file.txt  
sudo chown username:groupname file.txt  
sudo chown user:group file.txt
```

chgrp

Purpose:

Changes the group ownership of a file or directory.

```
sudo chgrp groupname file.txt
```

umask

Purpose:

Controls the default permissions assigned to newly created files and directories.

```
umask
umask 027
```

5. Process Management

ps

Purpose:

Displays information about running processes.

```
ps ps
aux
ps aux | grep nginx
```

top

Purpose:

Displays running processes and real-time CPU and memory usage.

```
top
```

Press Q to exit.

htop

Purpose:

Provides an interactive view of running processes.

```
htop
```

pgrep

Purpose:

Finds the process ID(s) of a process based on its name or other attributes.

```
pgrep nginx
```

pidof

Purpose:

Displays the process ID of a running program.

```
pidof nginx
```

kill

Purpose:

Sends a signal to a running process.

```
kill PID
kill -9 PID
```

pkill

Purpose:

Terminates (or signals) processes based on their name.

```
pkill nginx
```

nice

Purpose:

Starts a process with a specified CPU scheduling priority.

```
nice -n 10 command
```

renice

Purpose:

Changes the priority of an already running process.

```
renice 10 -p PID
```

jobs

Purpose:

Displays jobs running in the current shell.

```
jobs
```

bg

Purpose:

Resumes a stopped job in the background.

```
bg
```

fg

Purpose:

Brings a background job to the foreground.

```
fg
```

6. System Services (systemd)

systemctl

Purpose:

Controls and manages systemd services.

```
systemctl status nginx sudo
systemctl start nginx sudo
systemctl stop nginx sudo
systemctl restart nginx sudo
systemctl reload nginx sudo
systemctl enable nginx sudo
systemctl disable nginx
```

systemctl list-units

Purpose:

Lists currently loaded systemd units.

```
systemctl list-units
```

systemctl list-unit-files

Purpose:

Lists installed systemd service unit files and their enabled/disabled state.

```
systemctl list-unit-files
```

journalctl

Purpose:

Views logs collected by the systemd journal.

```
journalctl
journalctl -u nginx
journalctl -n 50
journalctl -f
journalctl -b
```

journalctl -p

Purpose:

Displays journal entries filtered by priority.

```
journalctl -p err
```

journalctl --since

Purpose:

Displays logs from a specified time.

```
journalctl --since "1 hour ago"
```

dmesg

Purpose:

Displays messages from the Linux kernel ring buffer.

```
dmesg
```

7. Package Management

apt update

Purpose:

Updates the local package index.

```
sudo apt update
```

apt upgrade

Purpose:

Upgrades installed packages.

```
sudo apt upgrade
```

apt install

Purpose:

Installs a software package.

```
sudo apt install nginx
```

apt remove

Purpose:

Removes an installed software package.

```
sudo apt remove nginx
```

apt purge

Purpose:

Removes a package and its configuration files where applicable.

```
sudo apt purge nginx
```

apt search

Purpose:

Searches for available software packages.

```
apt search nginx
```

apt show

Purpose:

Displays detailed information about a package.

```
apt show nginx
```

apt-cache

Purpose:

Queries information about packages available through APT.

```
apt-cache policy nginx
```

dpkg

Purpose:

Manages Debian packages and displays installed package information.

```
dpkg -l | dpkg -l |  
grep nginx
```

8. Disk, Memory & System Information

df

Purpose:

Displays disk-space usage of mounted filesystems.

```
df -h
```

du

Purpose:

Displays disk usage of files and directories.

```
du -sh /var/log
```

lsblk

Purpose:

Displays information about block devices and storage devices.

```
lsblk
```

free

Purpose:

Displays RAM and swap memory usage.

```
free -h
```

vmstat

Purpose:

Displays information about processes, memory, CPU, and system performance.

```
vmstat
```

iostat

Purpose:

Displays CPU and disk input/output statistics.

```
iostat
```

Note: May require the sysstat package.

uptime

Purpose:

Displays how long the system has been running and the system load average.

```
uptime
```

uname

Purpose:

Displays Linux kernel and system information.

```
uname -a
```

hostnamectl

Purpose:

Displays information about the system hostname, operating system, kernel, and architecture.

```
hostnamectl
```

hostname

Purpose:

Displays the system's hostname.

```
hostname
```

timedatectl

Purpose:

Displays and manages system date, time, and timezone information.

```
timedatectl
```

date

Purpose:

Displays the current system date and time.

```
date
```

lscpu

Purpose:

Displays information about the system CPU.

```
lscpu
```

lsusb

Purpose:

Displays connected USB devices.

```
lsusb
```

lspci

Purpose:

Displays information about PCI devices.

```
lspci
```

lsof

Purpose:

Lists open files and the processes using them.

```
sudo lsof sudo  
lsof -i :80
```

9. Networking

ip

Purpose:

Displays and manages network interfaces, IP addresses, and routing information.

```
ip addr  
ip link  
ip route
```

ping

Purpose:

Tests basic network connectivity to a host.

```
ping 8.8.8.8  
ping google.com
```

ss

Purpose:

Displays network connections, listening ports, and socket information.

```
ss -tuln sudo  
ss -tulpn
```

arp

Purpose:

Displays or manipulates the ARP cache on systems where the command is available.

```
arp -a
```

curl

Purpose:

Transfers data to or from a server; commonly used to test HTTP/HTTPS services.

```
curl http://localhost  
curl -I http://localhost  
curl -v http://localhost
```

wget

Purpose:

Downloads files from HTTP, HTTPS, and other supported protocols.

```
wget https://example.com/file.zip
```

nc

Purpose:

Tests TCP or UDP network connections.

```
nc -zv hostname 80
```

dig

Purpose:

Performs DNS queries and displays DNS information.

```
dig example.com
dig example.com A
```

nslookup

Purpose:

Queries DNS information.

```
nslookup example.com
```

host

Purpose:

Performs simple DNS lookups.

```
host example.com
```

getent

Purpose:

Queries system databases such as DNS and user information.

```
getent hosts example.com
getent passwd username
```

traceroute

Purpose:

Shows the network path and hops between the local machine and a destination.

```
traceroute example.com
```

mtr

Purpose:

Combines ping and traceroute functionality to help diagnose network problems.

```
mtr google.com
```

Note: May need to be installed separately.

resolvectl

Purpose:

Displays DNS resolver information and status on systems using systemd-resolved.

```
resolvectl status
```

iptables

Purpose:

Manages firewall rules on systems using iptables.

```
sudo iptables -L
```

Note: Modern distributions may use nftables or another firewall system instead.

ufw

Purpose:

Provides a simplified interface for managing firewall rules on Ubuntu/Debian-based systems.

```
sudo ufw status sudo
ufw allow 22/tcp sudo
ufw deny 23/tcp
```

10. SSH & Remote Administration

ssh

Purpose:

Connects securely to a remote Linux system.

```
ssh username@server_ip ssh -i
key.pem username@server_ip
```

ssh-keygen

Purpose:

Generates SSH public and private key pairs.

```
ssh-keygen ssh-keygen
-t ed25519
```

ssh-copy-id

Purpose:

Copies an SSH public key to a remote server for key-based authentication.

```
ssh-copy-id username@server_ip
```

scp

Purpose:

Securely copies files between systems using SSH.

```
scp file.txt username@server_ip:/path/
scp username@server_ip:/path/file.txt .
```

sftp

Purpose:

Transfers files securely using SSH.

```
sftp username@server_ip
```

11. Archiving & Compression

tar

Purpose:

Creates and extracts archive files.

```
tar -cvf archive.tar directory/ tar
-xvf archive.tar tar -czvf
archive.tar.gz directory/ tar -xzvf
archive.tar.gz
```

gzip

Purpose:

Compresses files using gzip compression.

```
gzip file.txt
```

gunzip

Purpose:

Decompresses gzip-compressed files.

```
gunzip file.txt.gz
```

12. Environment, Shell & History

echo

Purpose:

Displays text or the value of a variable.

```
echo "Hello Linux"  
echo $HOME
```

env

Purpose:

Displays environment variables, or runs a command with modified environment variables.

```
env
```

printenv

Purpose:

Displays the value of an environment variable.

```
printenv HOME
```

export

Purpose:

Creates or modifies an environment variable for the current shell and its child processes.

```
export NAME=Shayan
```

history

Purpose:

Displays previously executed shell commands.

```
history
```

clear

Purpose:

Clears the terminal screen.

```
clear
```

man

Purpose:

Displays the manual page for a command.

```
man chmod man  
systemctl
```

which

Purpose:

Displays the location of an executable command.

```
which python
which nginx
```

type**Purpose:**

Shows how the shell interprets a command.

```
type cd
type ls
```

command**Purpose:**

Checks or executes a command without triggering certain shell aliases or functions.

```
command -v nginx
```

source**Purpose:**

Executes commands from a file in the current shell environment.

```
source ~/.bashrc
```

sleep**Purpose:**

Pauses execution for a specified amount of time.

```
sleep 5
```

exit**Purpose:**

Terminates the current shell or script with a specified exit status.

```
exit 0
```

13. Bash Scripting & Automation

bash**Purpose:**

Runs commands and Bash scripts using the Bash shell.

```
bash script.sh
```

chmod +x**Purpose:**

Makes a script executable.

```
chmod +x script.sh
```

./script.sh**Purpose:**

Executes an executable script from the current directory.

```
./script.sh
```

read

Purpose:

Reads input from the user or standard input into a variable.

```
read name
echo "$name"
```

test / [

Purpose:

Evaluates conditions in shell scripts.

```
test -f file.txt
[ -f file.txt ]
```

\$?

Purpose:

Stores the exit status of the most recently executed command; used to check success/failure.

```
ls
echo
$?

command
if [ $? -eq 0 ];
then echo "Success"
else echo "Failed"
fi
```

\$0

Purpose:

Represents the name of the currently running Bash script.

```
echo "$0"
```

\$1

Purpose:

Represents the first positional argument passed to a Bash script.

```
./script.sh hello

# Inside the script: echo "$1"
```

\$#

Purpose:

Displays the number of positional arguments passed to a Bash script.

```
echo "$#"
```

\$@

Purpose:

Represents all positional arguments passed to a Bash script.

```
echo "$@"
```

crontab

Purpose:

Schedules commands or scripts to run automatically at specified times.

```
crontab -e
```

14. Shell Operators & Redirection

>

Purpose:

Redirects command output to a file and overwrites the existing content.

```
echo "Hello" > file.txt
```

>>

Purpose:

Appends command output to the end of a file.

```
echo "New line" >> file.txt
```

|

Purpose:

Passes the output of one command as input to another command.

```
ps aux | grep nginx
```

&&

Purpose:

Runs the second command only when the first command succeeds.

```
mkdir project && cd project
```

||

Purpose:

Runs the second command when the first command fails.

```
ping server || echo "Connection failed"
```

15. Nginx & Web Server

nginx -t

Purpose:

Tests the Nginx configuration for syntax errors.

```
sudo nginx -t
```

nginx -v

Purpose:

Displays the installed Nginx version.

```
nginx -v
```

systemctl reload nginx

Purpose:

Reloads the Nginx configuration without completely stopping the service.

```
sudo systemctl reload nginx
```

curl -I

Purpose:

Displays HTTP response headers and helps verify web-server behavior.

```
curl -I http://localhost
```

curl -v

Purpose:

Displays detailed information about an HTTP/HTTPS connection for troubleshooting.

```
curl -v http://localhost
```

tail -f

Purpose:

Continuously monitors a log file for new entries.

```
sudo tail -f /var/log/nginx/access.log
```

grep (logs)

Purpose:

Searches Nginx logs for specific events, errors, or patterns.

```
sudo grep "404" /var/log/nginx/access.log
```

16. Important Linux Concepts

File Permissions

Controls who can read, write, or execute a file or directory.

Permissions: r = Read, w = Write, x = Execute.

Permission categories: User/Owner, Group, Others.

Least Privilege

Ensures that users and processes receive only the permissions required to perform their tasks. For example, a normal user should not be given unnecessary root or administrative privileges.

Root User

The root user has extensive administrative privileges on a Linux system. Administrative commands can commonly be executed using: sudo command

Exit Status

Indicates whether a command completed successfully. Checked via: echo \$?

Generally: 0 = Success, Non-zero = Failure.